

## 7. GOVERNANCE, LINEAGE & RELEASE GATE

STPA Hazard Mitigation · Bumpless Human Joystick Override · Defensible Release Case (**LOOP-01** → **REL-01**)

### 1. SENSING

Photons / Voltages to DMA  
MIPI CSI-2 · I2C · SPI Bus Priority

### 2. PERCEPTION

Encoders to Spatial Tokens  
ViT · DINOv2 · 3D Affordances

### 3. WORLD MODELS

Latent Belief &  $SE(3)$  Trees  
JEPA / RSSM Dynamics · TTL Validity

SYSTEM 1.5 · LINUX MPU / NPU

### 5. TRAJECTORY DECODERS

Diffusion Policies · ACT Action Chunking ( $H$ -steps) · Jerk Bounds

SYSTEM 2 · LINUX MPU

### 4. SEMANTIC DELIBERATION

VLMs & Spatial Foundation Models · Goals · Leases ( $t_{\text{expire}}$ )

Expiring Proposal  $p_t$

THE PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (NO DIRECT MOTOR ACCESS)

SYSTEM 1 · REAL-TIME MCU (BARE-METAL / FreeRTOS)

### 6. REAL-TIME REFLEX & SAFETY ENFORCEMENT

1 kHz Reflex Timing Loop · Control Barrier Functions ( $h(x) \geq 0$ ) · Stopping Distance  $d_{\text{stop}}(v_t) \leq d_{\text{clearance}}$  · Veto ( $u_t$ )

Permitted Action  $u_t = \text{permit}(p_t)$

THE PHYSICAL WORLD ( $W_t \rightarrow W_{t+1}$ )

Kinetic Momentum ( $p=mv$ ) · Joule Heat Dissipation ( $I^2R$ ) · Matter Mutation · Friction ( $\mu$ ) · Collision Dynamics

Endogenous Sensory Shift ( $A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$ )